PIPELINE COMMAND CENTER

Midstream Pipeline Digital Twin & Agentic AI

DATABRICKS SOLUTION ACCELERATOR

Powered by Databricks Apps • Unity Catalog • Serverless

REAL-TIME DIGITAL TWIN & AGENTIC AI FOR MIDSTREAM PIPELINE OPERATIONS

Pipeline Command Center delivers a unified command center for the Eagle Ford Midstream Trunk — fusing live telemetry, predictive maintenance, AI crew allocation, leak detection, and compliance monitoring across 18 assets spanning 87.3 miles.

◆ BUSINESS VALUE

- **Detect pipeline leaks early** with dedicated AI leak detection agent analyzing pressure differentials
- **Optimize crew dispatch** with skill-based allocation, proximity scoring, and reasoning chains
- **Predict asset failures** with RUL forecasting for pipe segments, compressors, valves, and cathodic protection
- **Ensure regulatory compliance** with continuous monitoring, event logging, and audit-ready dashboards
- **Centralize pipeline operations** — telemetry, work orders, crew, and AI advisors in one platform

◆ CHALLENGES ADDRESSED

- **Leak detection delays** — manual monitoring missing time-critical pressure drop events
- **Reactive crew dispatch** — no intelligent assignment considering certifications and proximity
- **Siloed asset management** — compressor stations, valves, and CP systems monitored independently
- **Integrity blind spots** — wall thinning and corrosion detected only during scheduled pigging runs
- **Compliance gaps** — fragmented event logs making regulatory reporting difficult

◆ WHO IT'S FOR

🔧 **Pipeline Integrity Engineer**
Digital twin visualization, wall thickness monitoring, and RUL predictions

👷 **Operations Supervisor**
Real-time pipeline overview, crew dispatch, and event/alarm management

🛡️ **HSE / Compliance Officer**
Regulatory audit trails, leak detection alerts, and compliance dashboards

📊 **Asset Manager**
Predictive maintenance KPIs, spare parts inventory, and work order tracking

18	6	8	87.3mi
Assets Monitored	AI Sub-Agents	Dashboard Tabs	Pipeline Length

◆ QUICK START GUIDE

1. **Clone Repository**
   Clone the Pipeline Command Center repo to your local machine
2. **Import to Workspace**
   Use Databricks CLI to import app files into your workspace
3. **Deploy the App**
   Run databricks apps deploy to launch the command center
4. **Monitor & Explore**
   Open the app URL and explore all 8 tabs

◆ GETTING THE MOST OUT OF PIPELINE COMMAND CENTER

📡 **Watch the Digital Twin**
Use the full-width SVG schematic to monitor all 18 assets with traffic-light health

🤖 **Consult Pipeline Advisor**
Chat with the agentic AI for natural-language insights and recommendations

👥 **Review Crew Allocation**
See AI-driven crew dispatch with certification matching and proximity scoring

⚙️ **Track Predictive Maintenance**
Monitor RUL predictions and failure probabilities to plan interventions early

◆ TARGET INDUSTRY

Midstream Oil & Gas — Pipeline operations, gas gathering systems, crude oil trunk lines, and NGL transportation networks. Applicable to any operator under PHMSA/DOT regulations.

⚠ **DEMO ONLY** — This application is a demonstration and solution accelerator. Please refer to the included LICENSE file for full terms of use.

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